

Infinite realizability and parity of the Mostar index

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Abstract

A nonnegative integer is infinitely realizable by the Mostar index if infinitely many pairwise non-isomorphic connected graphs have that index. Alizadeh et al. proved that every nonnegative even integer is infinitely realizable and asked which odd integers have this property. We prove that every odd integer greater than one is infinitely realizable. Together with their results, this gives a complete characterization: a nonnegative integer is infinitely realizable if and only if it is different from one. Our constructions replace the vertices of an odd cycle by complete graphs, except at one or two positions occupied by independent pairs. A single two-parameter family realizes every odd integer at least five, and a second family realizes three. In particular, five is infinitely realizable by chemical graphs. We also prove that cycle expansions using only complete graphs cannot realize three or five, which explains the role of the independent pairs. Finally, we give a transmission formula for the parity of the Mostar index and derive consequences for Eulerian graphs, regular graphs, and uniformly random labelled graphs.

Keywords: Mostar index; inverse problem; infinite realizability; graph distance; parity.

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1 Introduction

The Mostar index, introduced by Došlić et al. [1], measures the distance imbalance across the edges of a graph. It is a bond-additive invariant in chemical graph theory; see Ali and Došlić [2] for a survey. Throughout this paper, graphs are finite, nonempty, simple, undirected, and connected, unless a random graph is explicitly under consideration. For an edge uv of a graph G , let

$$n_u(uv) = |\{x \in V(G) : d_G(x, u) < d_G(x, v)\}|,$$

and define $n_v(uv)$ analogously. The Mostar index is

$$\text{Mo}(G) = \sum_{uv \in E(G)} |n_u(uv) - n_v(uv)|.$$

The transmission of a vertex v is $\text{Tr}_G(v) = \sum_{x \in V(G)} d_G(v, x)$. A *chemical graph* is a connected graph of maximum degree at most four.

The inverse problem asks which nonnegative integers occur as Mostar indices. Alizadeh et al. [3] proved that every nonnegative integer except one occurs. They also proved that every nonnegative even integer is infinitely realizable [3, Proposition 3]. Their Problem 1 asks

whether any odd integer is infinitely realizable, whether every odd integer greater than one is infinitely realizable, and, if not, which ones are. Here *infinitely realizable* always refers to infinitely many pairwise non-isomorphic connected simple graphs.

We answer this problem completely.

Theorem 1.1. *A nonnegative integer is infinitely realizable by the Mostar index if and only if it is different from one.*

The new part of Theorem 1.1 is the infinite realizability of every odd integer greater than one. It follows from two explicit constructions. The first is a family $H_{a,t}$, with integers $a \geq 1$ and $t \geq 0$, satisfying

$$\text{Mo}(H_{a,t}) = 2a + 3, \quad |V(H_{a,t})| = (a + 3)(2t + 3) + 2, \quad \Delta(H_{a,t}) = a + 3. \quad (1.1)$$

The second is a family G_t , with $t \geq 0$, satisfying

$$\text{Mo}(G_t) = 3, \quad |V(G_t)| = 10t + 19, \quad \Delta(G_t) = 5. \quad (1.2)$$

Taking $a = 1$ in (1.1) gives the following additional conclusion.

Corollary 1.2. *There are infinitely many pairwise non-isomorphic chemical graphs with Mostar index five.*

The distinction between ordinary and infinite realizability is essential. Finding one graph of index three or five does not answer Problem 1 of [3]. Likewise, Alex and Das [4] established eventual realizability for each fixed cyclomatic number $c \geq 3$, whereas the present work concerns infinitely many realizations of each fixed value. Our constructions have unbounded order while their Mostar index remains fixed.

The proof uses expansions of odd cycles. Each cycle vertex is replaced by a complete graph, and consecutive parts are joined completely. At one or two specified positions, we replace a complete graph of order two by two nonadjacent vertices. This changes just one within-part distance at each position. The resulting unit changes in transmission remove unwanted edge contributions. We prove in Section 5 that expansions using only complete parts cannot have index three or five. Thus these local changes overcome an obstruction to the more restricted construction.

Alizadeh et al. [3] also observed that even Mostar indices predominate in their small-order graph enumerations and asked about their parity. Related parity results for integer-valued topological indices were recently studied by Došlić [5]. In Section 6 we establish a degree–transmission congruence and deduce that Eulerian and regular graphs have even Mostar index. We also give an asymptotic statement for uniformly random *labelled* graphs. This probability statement is separate from a comparison of the numbers of unlabelled graphs of a fixed small order.

The paper is organized as follows. Section 2 gives the transmission formulas used throughout. Sections 3 and 4 construct the two families and prove Theorem 1.1. Section 5 establishes the obstruction for complete parts. Section 6 treats parity, and Section 7 records the scope of the results. All infinite families are proved by exact identities; no computational enumeration is used in the proofs.

2 Transmission calculus for cycle expansions

We use the standard identity

$$\text{Tr}_G(u) - \text{Tr}_G(v) = n_v(uv) - n_u(uv) \quad (uv \in E(G)). \quad (2.1)$$

Its transmission interpretation appears in Balakrishnan et al. [6]; the corresponding expression for the Mostar index is recorded by Alizadeh et al. [7, Corollary 2.2]. In particular,

$$\text{Mo}(G) = \sum_{uv \in E(G)} |\text{Tr}_G(u) - \text{Tr}_G(v)|. \quad (2.2)$$

We invoke these identities as known results.

Let $N \geq 3$, and let $\mathbf{s} = (s_0, \dots, s_{N-1})$ be a sequence of positive integers. Let $I \subseteq \{0, \dots, N-1\}$. The graph $X(\mathbf{s}; I)$ has disjoint parts $X_0, \dots, X_{N-1}$ of sizes $s_0, \dots, s_{N-1}$. A part X_i induces an independent set when $i \in I$, and a complete graph otherwise. Every vertex of X_i is adjacent to every vertex of X_{i+1} , with indices modulo N . There are no other edges. Write $X(\mathbf{s}) = X(\mathbf{s}; \emptyset)$. We refer to the parts as *levels*. Vertices in the same level have equal transmission.

Let d_N denote distance on the cycle C_N , and put

$$e_i = \begin{cases} s_i - 1, & i \in I, \\ 0, & i \notin I. \end{cases}$$

Lemma 2.1. *The transmission T_i of each vertex in level X_i of $X(\mathbf{s}; I)$ satisfies*

$$T_i = \sum_{k=0}^{N-1} d_N(i, k) s_k + s_i - 1 + e_i. \quad (2.3)$$

Moreover,

$$\text{Mo}(X(\mathbf{s}; I)) = \sum_{i=0}^{N-1} s_i s_{i+1} |T_{i+1} - T_i|. \quad (2.4)$$

If $N = 2h + 1$, and $D_i = T_{i+1}^{\text{cl}} - T_i^{\text{cl}}$ denotes the transmission difference for $X(\mathbf{s})$, then

$$D_i - D_{i-1} = s_{i-1} + s_{i+1} - s_{i+h} - s_{i+h+1}. \quad (2.5)$$

Proof. For distinct levels i, k , the distance between any vertex of X_i and any vertex of X_k is $d_N(i, k)$. Indeed, a cycle path gives a path of that length, and any graph path projects to a walk on the cycle after steps inside a level are suppressed. Within a level, the distance between distinct vertices is one for a complete level and two for an independent level. This proves (2.3). Edges inside a level have zero contribution to (2.2); each of the $s_i s_{i+1}$ edges between consecutive levels has contribution $|T_{i+1} - T_i|$. This proves (2.4).

For an odd cycle, its distance function has second difference

$$d_N(i+1, k) - 2d_N(i, k) + d_N(i-1, k) = 2\mathbf{1}_{\{k=i\}} - \mathbf{1}_{\{k=i+h\}} - \mathbf{1}_{\{k=i+h+1\}}.$$

Apply this identity to (2.3) with $e_i = 0$. The second difference of $s_i - 1$ cancels the term $2s_i$, giving (2.5). $\square$

We write $(b_1, \dots, b_q)^t$ for t consecutive copies of a block, with the empty block understood when $t = 0$. In a displayed level sequence, 2_1 indicates an independent pair; every other entry indicates a complete level.

3 A uniform family for every odd value at least five

Fix integers $a \geq 1$ and $t \geq 0$, and put

$$h = 3t + 4, \quad N = 2h + 1 = 6t + 9, \quad \mathbf{q} = (a, 1, 2)^{2t+3}.$$

Define the level sizes by

$$s_i = q_i + \mathbf{1}_{\{i=h\}} + \mathbf{1}_{\{i=2h\}},$$

and let

$$H_{a,t} = X(\mathbf{s}; \{h+1\}). \quad (3.1)$$

Since $h \equiv 1 \pmod{3}$, the enlarged levels have sizes $s_h = 2$ and $s_{2h} = 3$, and level $h+1$ is an independent pair. Equivalently,

$$H_{a,t} = X((a, 1, 2)^t, a, 1, 2, a, 2, 2_I, (a, 1, 2)^t, a, 1, 3).$$

Here and below the notation with 2_I includes the specified independent levels.

Theorem 3.1. *For every $a \geq 1$ and $t \geq 0$,*

$$\text{Mo}(H_{a,t}) = 2a + 3.$$

The order, size, and maximum degree are

$$|V(H_{a,t})| = (a+3)(2t+3) + 2, \quad (3.2)$$

$$|E(H_{a,t})| = (2t+3) \frac{(a+2)(a+3)}{2} + 2a + 5, \quad (3.3)$$

$$\Delta(H_{a,t}) = a + 3. \quad (3.4)$$

Proof. First consider $X(\mathbf{q})$, with every level complete. Because $\mathbf{q}$ has period three and $h \equiv 1 \pmod{3}$, the right-hand side of (2.5) is zero at every position. Its cyclic transmission differences are therefore constant and sum to zero, so every difference is zero. All levels of $X(\mathbf{q})$ have the same transmission, say B . Summing (2.3) over the N levels gives

$$NB = (h(h+1) + 1) \sum_i q_i - N,$$

since the sum of distances from a vertex of C_{2h+1} is $h(h+1)$. As $\sum_i q_i = N(a+3)/3$, it follows that

$$B = (a+3)(3t^2 + 9t + 7) - 1. \quad (3.5)$$

By (2.3), the transmissions in $H_{a,t}$ are

$$\begin{aligned} T_i &= B + d_N(i, h) + d_N(i, 2h) \\ &\quad + \mathbf{1}_{\{i=h\}} + \mathbf{1}_{\{i=2h\}} + \mathbf{1}_{\{i=h+1\}}. \end{aligned} \quad (3.6)$$

The shorter h - $2h$ arc consists of levels $h, h+1, \dots, 2h$ and has length h ; its complementary arc has length $h+1$. On the shorter arc the sum of the two distances in (3.6) is h . At every interior vertex of the complementary arc that sum is $h+1$. Hence

$$T_i = \begin{cases} B + h, & h+2 \leq i \leq 2h-1, \\ B + h + 1, & \text{otherwise.} \end{cases} \quad (3.7)$$

There are exactly two nonzero consecutive differences:

$$T_{h+2} - T_{h+1} = -1, \quad T_{2h} - T_{2h-1} = 1.$$

Their interface sizes are

$$s_{h+1}s_{h+2} = 2a, \quad s_{2h-1}s_{2h} = 3.$$

Equation (2.4) therefore gives $\text{Mo}(H_{a,t}) = 2a + 3$.

The order formula follows by adding the level sizes. In the baseline $X(\mathbf{q})$, each period contributes $\binom{a}{2} + 1 + 3a + 2 = (a+2)(a+3)/2$ edges. The two enlargements each add $a+3$ edges, and making level $h+1$ independent removes one edge. This proves (3.3). Finally, a vertex in a complete level i has degree $s_{i-1} + s_i + s_{i+1} - 1$; in an independent level its degree is $s_{i-1} + s_{i+1}$. These expressions are at most $a+3$ here, with equality at level 0, proving (3.4). $\square$

For any fixed odd $p \geq 5$, take $a = (p - 3)/2$ and let t vary. The orders in (3.2) strictly increase, proving infinite realizability. With $a = 1$, the maximum degree is four, which proves Corollary 1.2.

4 An infinite family with Mostar index three

Fix $t \geq 0$, and put

$$N = 6t + 11, \quad j = 3t + 2, \quad h = 3t + 5.$$

Define

$$G_t = X(1, 3, (1, 1, 3)^t, 1, 2, 2_1, 2_1, (1, 1, 3)^t, 1, 1, 3, 2, 1). \quad (4.1)$$

The independent levels are $j + 2$ and $j + 3$.

Theorem 4.1. *For every $t \geq 0$,*

$$\text{Mo}(G_t) = 3, \quad |V(G_t)| = 10t + 19, \quad |E(G_t)| = 20t + 39, \quad \Delta(G_t) = 5.$$

Proof. Let $\mathbf{s}$ be the level-size sequence in (4.1). First make every level complete, and let $D_i = T_{i+1}^{\text{cl}} - T_i^{\text{cl}}$. Set $R_i = D_i - D_{i-1}$. Equation (2.5) gives

$$R_i = s_{i-1} + s_{i+1} - s_{i+h} - s_{i+h+1}.$$

Table 1 verifies this expression at every position. All indices in the four entries are taken modulo N . A row with an empty index range contributes nothing. In particular, the table is valid when $t = 0$, and the seven positions with nonzero R_i are distinct.

Table 1: An exact difference certificate for the complete-level baseline of G_t . The residue conditions refer to i modulo three.

Position or range	Residue	$(s_{i-1}, s_{i+1}, s_{i+h}, s_{i+h+1})$	R_i
0		(1, 3, 2, 1)	1
j		(3, 2, 1, 3)	1
$j + 1$		(1, 2, 3, 2)	-2
$j + 2$		(2, 2, 2, 1)	1
$j + 3$		(2, 1, 1, 1)	1
$j + 4$		(2, 1, 1, 3)	-1
$N - 1$		(2, 1, 2, 2)	-1
$1 \leq i \leq j - 1$	0	(1, 3, 3, 1)	0
	1	(1, 1, 1, 1)	0
	2	(3, 1, 1, 3)	0
$j + 5 \leq i \leq N - 4$	0	(3, 1, 1, 3)	0
	1	(1, 3, 3, 1)	0
	2	(1, 1, 1, 1)	0
$N - 3$		(1, 2, 1, 2)	0
$N - 2$		(3, 1, 2, 2)	0

The unique cyclic sequence with these differences and $\sum_i D_i = 0$ is

$$D_j = D_{j+3} = 1, \quad D_{j+1} = D_{N-1} = -1, \quad (4.2)$$

with all other $D_i = 0$. To see uniqueness, any two such sequences have identical first differences, so their difference is constant; the zero-sum condition makes that constant zero.

Now delete the internal edge in each of levels $j + 2, j + 3$. By (2.3), the transmissions in these two levels each increase by one, and the other transmissions do not change. Consequently, the differences at $j + 1$ and $j + 3$ in (4.2) vanish. The final graph has exactly

$$T_{j+1} - T_j = 1, \quad T_0 - T_{N-1} = -1$$

as its nonzero consecutive differences. Their interface sizes are $s_j s_{j+1} = 2$ and $s_{N-1} s_0 = 1$. Equation (2.4) gives $\text{Mo}(G_t) = 2 + 1 = 3$.

Adding the level sizes gives $10t + 19$ vertices. The degree formula used in Theorem 3.1 gives four vertices of degree three, $10t + 9$ vertices of degree four, and six vertices of degree five. Thus the maximum degree is five, and the degree sum is $40t + 78$, giving $20t + 39$ edges. $\square$

For completeness, the full transmission profile is

$$T_i = \begin{cases} C_t, & 0 \leq i \leq j, \\ C_t + 1, & j + 1 \leq i \leq N - 1, \end{cases} \quad C_t = 15t^2 + 56t + 52. \quad (4.3)$$

The differences were proved above. The initial value follows directly from (2.3), since neither independent level contains a vertex of level zero:

$$\begin{aligned} T_0 &= 3 + \sum_{r=0}^{t-1} (15r + 17) + (21t + 26) \\ &\quad + \sum_{r=0}^{t-1} (15t + 18 - 15r) + 23 = 15t^2 + 56t + 52. \end{aligned}$$

The two transmission classes have sizes $5t + 5$ and $5t + 14$, respectively.

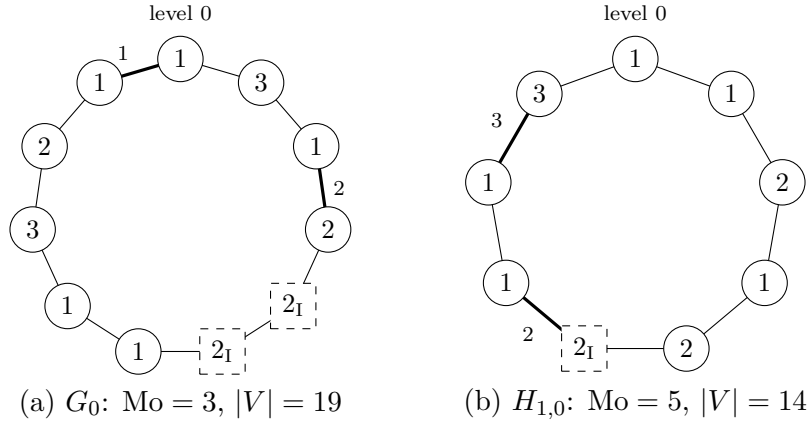


Figure 1: Level diagrams of the first members of the two families. A circle labelled s represents a complete graph of order s ; a dashed box represents an independent pair. Each drawn cycle edge represents a complete bipartite join of its two levels. Bold interfaces are the only ones with nonzero transmission difference; their labels give their total Mostar contributions. Levels are numbered clockwise from the top.

Proof of Theorem 1.1. The value one is not realizable, and every nonnegative even integer is infinitely realizable, by Alizadeh et al. [3]. For an odd integer $p \geq 5$, fix $a = (p - 3)/2$ and apply Theorem 3.1. Theorem 4.1 treats $p = 3$. In both cases the graph order strictly increases with t , so the graphs are pairwise non-isomorphic. These cases exhaust $\mathbb{N}_0 \setminus \{1\}$. $\square$

5 An obstruction when every level is complete

Allowing noncomplete levels is necessary to realize three or five by cycle expansions. This follows from a cyclic identity that holds for all positive level sizes.

Proposition 5.1. *For every $N \geq 3$ and every positive integer sequence $\mathbf{s}$,*

$$\text{Mo}(X(\mathbf{s})) \notin \{3, 5\}.$$

Proof. Let T_i be the level transmissions and set $D_i = T_{i+1} - T_i$. We have

$$\sum_i D_i = 0, \quad \sum_i (s_i + s_{i+1}) D_i = 0. \quad (5.1)$$

The first identity telescopes. For the second, let C be the distance matrix of C_N and put $A = C + I_N$, where I_N is the identity matrix. In column-vector notation, (2.3) gives $\mathbf{T} = A\mathbf{s} - \mathbf{1}$. The matrix A is symmetric and circulant. It therefore commutes with the cyclic shift P , defined by $(P\mathbf{x})_i = x_{i+1}$. Consequently, $(P - P^{-1})A$ is skew-symmetric, and

$$\begin{aligned} \sum_i (s_i + s_{i+1})(T_{i+1} - T_i) &= \sum_i (s_i T_{i+1} - s_{i+1} T_i) \\ &= \mathbf{s}^\top (P - P^{-1}) A \mathbf{s} = 0. \end{aligned}$$

This proves (5.1).

Suppose that $\text{Mo}(X(\mathbf{s}))$ is three or five, and put $U = \sum_i |D_i|$. Since the integer differences sum to zero, U is a positive even integer. By (2.4), $U \leq \text{Mo}(X(\mathbf{s}))$, so $U \in \{2, 4\}$.

If $U = 2$, exactly two differences are nonzero, one equal to 1 and the other to -1 . The second identity in (5.1) forces the sums of the two endpoint level sizes at these interfaces to be equal, say to S . If $S = 2$, both interface products are one, and the Mostar index is two. If $S = 3$, both products are two, and the index is four. If $S \geq 4$, each product is at least $S - 1 \geq 3$, so the index is at least six. None is possible.

If $U = 4$, the Mostar index must be five. Then

$$\sum_i (s_i s_{i+1} - 1) |D_i| = 1.$$

Thus exactly one active interface has product two and a difference of absolute value one. Every other active interface has product one. Their endpoint-size sums are therefore three at the exceptional interface and two at all the others. The second sum in (5.1) is $2 \sum_i D_i + 1$ or $2 \sum_i D_i - 1$, a contradiction. $\square$

6 Parity and labelled graphs

We next record consequences of the transmission expression (2.2). They do not enter the proof of infinite realizability.

Proposition 6.1. *For every connected graph G ,*

$$\text{Mo}(G) \equiv \sum_{v \in V(G)} \deg_G(v) \text{Tr}_G(v) \equiv \sum_{\deg_G(v) \text{ odd}} \text{Tr}_G(v) \pmod{2}. \quad (6.1)$$

Proof. For integers x, y , $|x - y| \equiv x + y \pmod{2}$. Applying this to (2.2) and collecting the terms incident with each vertex proves the first congruence. Vertices of even degree contribute zero modulo two, giving the second. $\square$

Corollary 6.2. *Every Eulerian graph has even Mostar index. Every regular graph also has even Mostar index.*

Proof. In an Eulerian graph every degree is even, so (6.1) applies. If G is r -regular, then

$$\text{Mo}(G) \equiv r \sum_v \text{Tr}_G(v) = 2r \sum_{\{u,v\} \subseteq V(G)} d_G(u,v) \equiv 0 \pmod{2}.$$

□

Dehgardi and Azari [8, Theorem 5] proved that, for graphs of diameter at most two, the Mostar index equals degree irregularity; see also [7, Theorem 4.1]. Indeed, if $n = |V(G)|$ and $\text{diam}(G) \leq 2$, then $\text{Tr}_G(v) = 2(n-1) - \deg_G(v)$. Thus in this case the Mostar index equals $\sum_{uv \in E(G)} |\deg_G(u) - \deg_G(v)|$, the usual degree irregularity. Its parity is even; see also the discussion of degree-based indices in [5]. Directly from (6.1),

$$\text{Mo}(G) \equiv \sum_v \deg_G(v)^2 \equiv \sum_v \deg_G(v) = 2|E(G)| \equiv 0 \pmod{2}.$$

For a random graph, we state the event with connectedness explicitly so that no definition of transmission for disconnected graphs is needed.

Corollary 6.3. *If G is chosen from $G(n, 1/2)$, then, for $n \geq 2$,*

$$\Pr(G \text{ is connected and } \text{Mo}(G) \text{ is even}) \geq 1 - \frac{1}{2} \binom{n}{2} \left(\frac{3}{4}\right)^{n-2}.$$

In particular, this probability tends to one as n tends to infinity.

Proof. For any fixed pair of vertices, the probability that they are nonadjacent and have no common neighbor is $\frac{1}{2}(3/4)^{n-2}$. The union bound shows that, outside an event of probability at most the stated error term, every pair has distance at most two. The graph is then connected and has even Mostar index. □

Since $G(n, 1/2)$ is uniform on all labelled graphs of order n , this proves labelled asymptotic predominance of even values. It also holds after conditioning on connectedness, whose probability tends to one. It does not assert an exact ratio in fixed-order enumerations of isomorphism classes.

7 Further remarks and verification

Theorems 3.1 and 4.1 settle the odd infinite-realizability problem. Their mechanism is local: the periodic cycle expansion has controlled transmission, and replacing one or two K_2 levels by independent pairs changes only the necessary differences. Proposition 5.1 explains why complete levels alone cannot treat the two smallest odd values.

The chemical-graph conclusion is deliberately limited to index five. The graphs G_t have maximum degree five, so they do not establish infinite realizability of index three within chemical graphs. Another direction is to determine all possible orders of graphs with a prescribed Mostar index. The orders in our constructions are sufficient orders, not a classification of all possible orders and not assertions of minimum order.

The supplementary Python program `verify_constructions.py` constructs ordinary unweighted graphs and computes every distance by breadth-first search. It checks the Mostar index both from the original closer-vertex definition and from transmission differences, and compares the vertex transmissions with (3.7) and (4.3). These finite checks are supplementary; the proofs above are independent of them.

Statements and declarations

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Author contributions. Both authors contributed to the study conception, mathematical arguments, verification, and manuscript preparation.

Data and code availability. The constructions and proofs are contained in the article. The supplementary material provides a dependency-free Python verification program and its machine-readable output. No external empirical dataset is used.

Use of generative AI. ChatGPT (OpenAI) was used to explore candidate constructions, assist with proof development and verification code, and revise the exposition. The authors take responsibility for the mathematical arguments, computations, references, and final content.

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